

The `Float` class in Java is a wrapper class for the primitive data type `float`. It is part of the `java.lang` package and provides methods to work with float values, such as parsing, converting, and comparing them. The `Float` class also provides constants for representing positive and negative infinity, as well as NaN (Not-a-Number) values.

## Key Features of Float Wrapper Class

1. **Immutability:** Instances of the `Float` class are immutable.
2. **Constants:** Provides constants like `MAX_VALUE`, `MIN_VALUE`, `POSITIVE_INFINITY`, `NEGATIVE_INFINITY`, and `NaN`.
3. **Static Methods:** Methods for parsing strings to floats, comparing floats, and converting floats to strings.

## Float Class Methods

Here are some commonly used methods in the `Float` class:

1. `byte byteValue()`
2. `int compareTo(Float anotherFloat)`

- 3. `static int compare(float f1, float f2)`
- 4. `double doubleValue()`
- 5. `boolean equals(Object obj)`
- 6. `static float parseFloat(String s)`
- 7. `float floatValue()`
- 8. `int hashCode()`
- 9. `int intValue()`
- 10. `boolean isInfinite()`
- 11. `static boolean isInfinite(float v)`
- 12. `boolean isNaN()`
- 13. `static boolean isNaN(float v)`
- 14. `long longValue()`
- 15. `static String toString(float f)`
- 16. `String toString()`
- 17. `static Float valueOf(String s)`
- 18. `static Float valueOf(float f)`

## **1. byte byteValue()**

Returns the value of this `Float` object as a byte.

```
public class FloatExample {
```

```
public static void main(String[]
args) {
    Float f = 123.45f;
    byte value = f.byteValue();
    System.out.println(value);    //
Output: 123
}
```

## Output:

123

## 2. int compareTo(Float anotherFloat)

Compares two Float objects numerically.

```
public class FloatExample {
    public static void main(String[]
args) {
        Float f1 = 123.45f;
        Float f2 = 543.21f;

        System.out.println(f1.compareTo(f2));
        // Output: -1
    }
}
```

```
}
```

## Output:

```
-1
```

### **3. static int compare(float f1, float f2)**

Compares two float values numerically.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
  
    System.out.println(Float.compare(123.4  
5f, 543.21f)); // Output: -1  
    }  
}
```

## Output:

```
-1
```

### **4. double doubleValue()**

Returns the value of this `Float` object as a double.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f = 123.45f;  
        double value =  
f.doubleValue();  
        System.out.println(value);    //  
Output: 123.45  
    }  
}
```

**Output:**

```
123.44999694824219
```

## 5. `boolean equals(Object obj)`

Compares this object to the specified object.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f1 = 123.45f;  
        Float f2 = 123.45f;
```

```
System.out.println(f1.equals(f2)); //  
Output: true  
    }  
}
```

## Output:

```
true
```

## 6. static float parseFloat(String s)

Parses the string argument as a float.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        float f =  
Float.parseFloat("123.45");  
        System.out.println(f); //  
Output: 123.45  
    }  
}
```

## Output:

123.45

## 7. float floatValue()

Returns the value of this Float object as a float.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f = 123.45f;  
        float value = f.floatValue();  
        System.out.println(value);    //  
Output: 123.45  
    }  
}
```

## Output:

123.45

## 8. int hashCode()

Returns a hash code for this Float object.

```
public class FloatExample {  
    public static void main(String[]  
args) {
```

```
Float f = 123.45f;

System.out.println(f.hashCode()); //
Output: a unique hash code
    }
}
```

### Output:

```
1123477094
```

## 9. `intValue()`

Returns the value of this `Float` object as an `int`.

```
public class FloatExample {
    public static void main(String[]
args) {
        Float f = 123.45f;
        int value = f.intValue();
        System.out.println(value); //
Output: 123
    }
}
```

### Output:



123

## **10. boolean isInfinite()**

Checks if the value of this `Float` object is infinitely large in magnitude.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f =  
Float.POSITIVE_INFINITY;  
  
System.out.println(f.isInfinite());  
// Output: true  
    }  
}
```

## **Output:**

true

## **11. static boolean isInfinite(float v)**

Checks if the specified `float` value is infinitely large in magnitude.

```
public class FloatExample {
    public static void main(String[]
args) {

System.out.println(Float.isInfinite(Fl
oat.POSITIVE_INFINITY)); // Output:
true
    }
}
```

## Output:

```
true
```

## 12. boolean isNaN()

Checks if the value of this `Float` object is a Not-a-Number (NaN) value.

```
public class FloatExample {
    public static void main(String[]
args) {
        Float f = Float.NaN;
        System.out.println(f.isNaN());
// Output: true
    }
}
```

```
}
```

## Output:

```
true
```

### 13. static boolean isNaN(float v)

Checks if the specified `float` value is a Not-a-Number (NaN) value.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
  
    System.out.println(Float.isNaN(Float.N  
aN)); // Output: true  
    }  
}
```

## Output:

```
true
```

### 14. long longValue()

Returns the value of this `Float` object as a `long`.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f = 123.45f;  
        long value = f.longValue();  
        System.out.println(value);    //  
Output: 123  
    }  
}
```

**Output:**

123

## **15. static String toString(float f)**

Returns a `String` object representing the specified `float`.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
  
        System.out.println(Float.toString(123.  
45f));    // Output: 123.45
```

```
    }  
}
```

## Output:

```
123.45
```

## 16. String toString()

Returns a `String` object representing this `Float`'s value.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f = 123.45f;  
  
        System.out.println(f.toString()); //  
Output: 123.45  
    }  
}
```

## Output:

```
123.45
```

## 17. static Float valueOf(String s)

Returns a `Float` object holding the value given by the specified `String`.

```
public class FloatExample {  
    public static void main(String[]  
args) {  
        Float f =  
Float.valueOf("123.45");  
        System.out.println(f);    //  
Output: 123.45  
    }  
}
```

### Output:

```
123.45
```

## 18. static Float valueOf(float f)

Returns a `Float` instance representing the specified `float` value.

```
public class FloatExample {  
    public static void main(String[]  
args) {
```

```
Float f =  
Float.valueOf(123.45f);  
System.out.println(f); //  
Output: 123.45  
    }  
}
```

## Output:

123.45